Radian is the angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle (Fig 1.7).

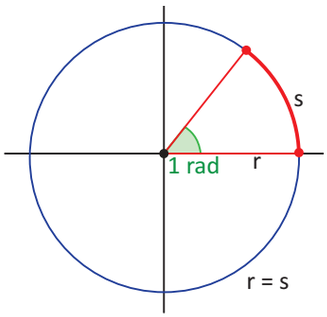



**Figure 1.7** Radian

$\pi$ radian = 180°

$$1 \text{ radian} = \frac{180°}{\pi}$$

**Problem 2**

Convert 60° into radian.

**Solution**

We know that,

$$1° = \frac{\pi}{180}$$

$$60° = \frac{\pi}{\cancel{180}\,3} \times \cancel{60} = \frac{\pi}{3} \text{ radian}$$

**Problem 3**

Convert $\frac{\pi}{4}$ into degrees.

**Solution**

We know that,

$\pi$ radian = 180°

$$\frac{\pi}{4} \text{ radian} = \frac{180}{4} = 45°$$

### 1.2.6 Solid angle

Solid angle is the angle formed by three or more planes intersecting at a common point. It can also be defined as 'angle formed at the vertex of the cone'. The SI unit of solid angle is 'steradian' and is denoted as 'sr'.

Steradian is the solid angle at the centre of a sphere subtended by a portion whose surface area is equal to the square of the radius of the sphere.

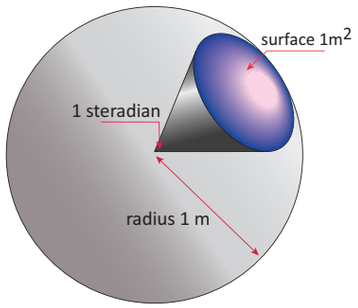



**Figure 1.8** Steradian

DO YOU KNOW?

Until 1995, plane angle and solid angle were classified under supplementary quantities. In 1995, they were shifted to derived quantities.

**Table 1.2** Difference between plane angle and solid angle

| Plane Angle | Solid Angle |
|---|---|
| It is the angle made at the point of intersection of two lines or planes. | It is the angle by the intersection of three or more planes at a common point. |
| It is two dimensional. | It is three dimensional. |
| Its unit is radian. | Its unit is steradian. |

## 1.3 Clocks

Clocks are used to measure time intervals. So many clocks are being used from the ancient time. Scientists have modified the mechanism of the clocks everytime to obtain accuracy.

### 1.3.1 Types of clock based on display

There are two types of clocks based on display. They are:

1. Analog clocks
2. Digital clocks

#### 1. Analog clocks

Analog clocks look like a classic clock. It has three hands to show the time.



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